Numerical integration of axisymmetric Helfrich shape equation

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March 12, 2026

Contents

1 Introduction

In this document I will explain how to use numerical methods to solve Helfrich shape equation. The method is general but I will show how to apply it to a specific system a nano particle interacting with a membrane.
This system has been inspired by the work of [1], the code is adapted from [4] and from supplementary materials of [3].
[1] volume is also conserved.

2 Previous work

The most useful resources on this are:

- Canalejo and liposki [2]
- [8] here they look at the dynamics. they approximate with a free energy depending on z only, the use shooting method and thet study the dynamics (quasi dynamics).
- [6] sensitivity analysis onto the vesicles shape equation, very mathy. they introduce also patch of spontaneous curvature.

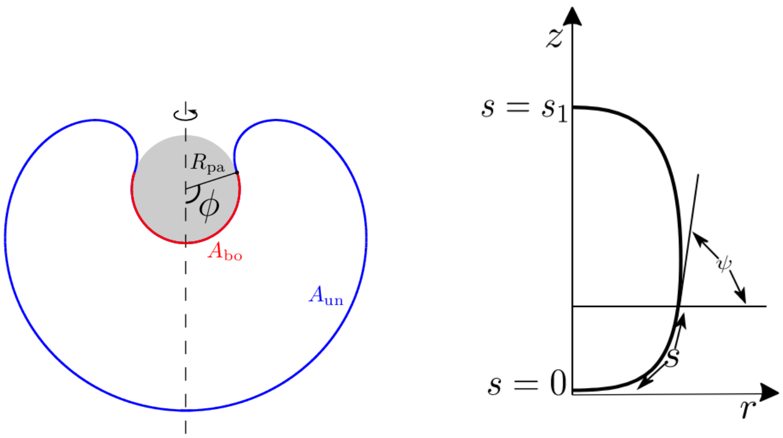
Worth mentioning:

- [5] this is apply same method to a plasma hole repair. lots of math in supplementary, it can be helpful.

3 Wrapping particle system

We are studying the interaction between a spherical membrane and a spherical particle. They interact with each other through an adhesive interaction. The spherical particle does not have any other properties. The membrane instead is modeled as an elastic surface following the Helfrich theory of elasticity. It has a bending modulus k and a surface tension Σ.

The system equilibrium state is the result of the interplay between two competing phenomena: the first one is the particle wrapping that decreases



(a) image taken from supplementary material of [1]

the free energy and the second one is the increase of energy due to the bending of the membrane surface.
The total free energy of the system will be composed by two contribution representing the bound and unbound segment:

$$E = E_{bo} + E_{un}$$

The bound segment of the membrane will follow the particle contour but the unbound segment does not have a trivial shape.
E_{bo} has an adhesive and a bending energy contribution: [1]

$$E_{bo} = (-2\pi|\bar{W}|r_{pa}^2 + 4r_{pa}^2)(1 - \cos\phi)$$

where ϕ is the wrapping angle and we are considering a vesicle bilayer with zero spontaneous curvature, ie $m = 0$.
In order to find the shape of the unbound segment that minimizes E_{un}, for a fixed value of contact angle ϕ, and satisfies the constraints on the total membrane area $A - A_{bo}$ and enclosed volume $V + V_{bo}$ of the vesicle, we must minimize the shape functional

$$F = E_{un} + \Sigma(A - A_{bo}) - \Delta P(V + V_{bo}) = \int_{A_{un}} dA\ 2kH^2 + \Sigma(A - A_{bo})$$

where $\int_{A_{un}} dA\ 2kH^2$ is the Helfrich energy integral.

Oss: In our particle based simulations we dont have any control or constraint on the vesicle volume, we dont pay a cost to change the vesicle volume then we put the volume term equal to zero.
Assuming that the vesicle shape will be axis-symmetric around z-axis It is possible to rewrite F in terms of $s, \psi(s), x(s)$:

$$F = \int_{s_0}^{s^\star} L(s, \psi(s), x(s))\, ds$$

The mean curvature H is given by

$$H = \frac{1}{2}(C_1 + C_2)$$

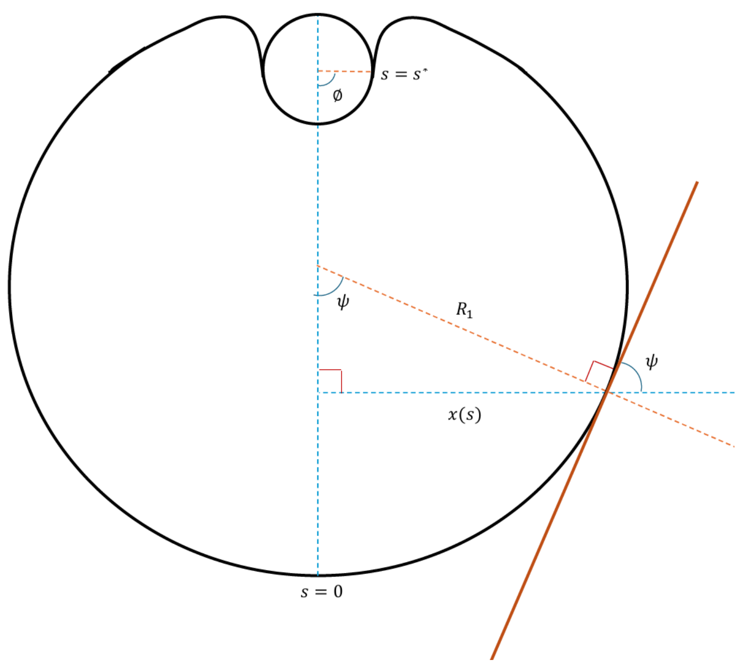


Figure 2: Visualization of the curvature

From figure 2, given that R_1 is literally the radius of curvature respect to the rotation axis, it follows that:

$$C_1 = \frac{1}{R_1} = \frac{\sin\psi}{x(s)}$$

C_2 is given by the definition of curvature: the rate at which ψ changes with respect to the arc length s, which gives:

$$C_2 = \frac{d\psi}{ds}$$

C_1 defines the curvature straight into (or out of) the paper, and C_2 defines the curvature in the direction of s.
These two curvatures together give the mean curvature:

$$H^2 = \frac{1}{4}\left(\frac{d\psi}{ds} + \frac{\sin\psi}{x(s)}\right)^2$$

Now we will parameterize F in terms of $s, \psi(s), x(s)$

$$F = \int_{A_{un}} dA\ 2kH^2 + \Sigma A_{un} = \int_{s_0}^{s^\star} L(s, \psi(s), x(s))\ ds$$

For this step we need to rewrite an integral in terms of dA to an integral in terms of s. To do this we use the following formula:

$$dA = 2\pi x ds$$

applying this formula to F gives:

$$F = \int_{A_{un}} dA\ 2kH^2 + \Sigma A_{un} = \int_{s_0}^{s^\star} 2\pi x ds\ 2kH^2 + \Sigma \int_{s_0}^{s^\star} 2\pi x ds$$

Now we will add the two integrals together and fill in H:

$$F = 2\pi\kappa \int_{s_0}^{s^\star} \frac{x}{2}\left(\dot{\psi} + \frac{\sin\psi}{x}\right)^2 + \frac{\Sigma}{\kappa} x ds$$

To make sure the relation between x and ψ is satisfied, the Lagrange multiplier $\gamma(s)$ is added to the integral.

$$F = 2\pi\kappa \int_{s_0}^{s^\star} \frac{x}{2}\left(\dot{\psi} + \frac{\sin\psi}{x}\right)^2 + \frac{\Sigma}{\kappa} x + \gamma \frac{\dot{x} - \cos\psi}{2\pi k} ds$$

$$L(s, \psi, \dot{\psi}, x, \dot{x}, \gamma) = \frac{x}{2}\left(\dot{\psi} + \frac{\sin\psi}{x}\right)^2 + \frac{\Sigma}{\kappa} x + \gamma \frac{\dot{x} - \cos\psi}{2\pi k}$$

$\gamma(s)$ is a Lagrange multiplier function that ensure the geometric relation between x and ψ is satisfied.
We aim to minimize the functional F using a variational approach, where $\delta F = 0$. Here, δF represents the variation with respect to the shape of the vesicle. Upon minimizing this functional, we obtain a system of differential equations that characterizes the shape of the vesicle.
This process is analogous to the method used in classical mechanics for minimizing the action functional with respect to a trajectory.

Euler-Lagrange equations: The solution of the minimization of functional F is given by the Euler-Lagrange equations.
In general, given a functional I:

$$I = \int_{x_a}^{x_b} L(x, f_1(x), f_2(x), \ldots, \dot{f}_1(x), \dot{f}_2(x), \ldots)$$

where $\dot{f}_i(x) = \frac{df_i}{dx}$ the Euler-Lagrange equations are:

$$\frac{d}{dx}\left[\frac{\partial L}{\partial \dot{f}_i}\right] - \frac{\partial L}{\partial f_i} = 0$$

The total variation δF is:

$$\delta F = \int_{s_1}^{s_2} \left[\left(\frac{d}{ds}\left[\frac{\partial L}{\partial \dot{\psi}}\right] - \frac{\partial L}{\partial \psi}\right)\delta\psi + \left(\frac{d}{ds}\left[\frac{\partial L}{\partial \dot{x}}\right] - \frac{\partial L}{\partial x}\right)\delta x\right] ds$$

The two Euler-Lagrange terms in equation must vanish separately at equilibrium, leading to the following differential equations:

$$\frac{d}{ds}\left[\frac{\partial L}{\partial \dot{\psi}}\right] - \frac{\partial L}{\partial \psi} = 0$$

Substituting will lead to

$$\ddot{\psi} = \frac{\sin\psi\cos\psi}{x^2} - \gamma\frac{\sin\psi}{2\pi\kappa x} - \frac{\dot{x}\dot{\psi}}{x}$$

Similarly, for the second term:

$$\frac{d}{ds}\left[\frac{\partial L}{\partial \dot{x}}\right] - \frac{\partial L}{\partial x} = 0$$

Substituting here gives:

$$\dot{\gamma} = 2\pi\Sigma + \kappa\pi\dot{\psi}^2 - \frac{k\pi\sin(\psi)^2}{x^2}$$

We can rewrite in first order terms substituing $\dot{\psi} = u$:

$$\begin{cases} \frac{d\psi}{ds} = u \\ \frac{du}{ds} = [-\frac{u}{x}\cos\psi + \frac{\cos(\psi)\sin(\psi)}{x^2} + \frac{\gamma\sin\psi}{2\pi\kappa x}] \\ \frac{d\gamma}{ds} = [\pi\kappa(u^2 - \frac{\sin^2\psi}{x^2}) + 2\pi\Sigma] \end{cases} \tag{1}$$

Hamiltonian function:

$$H = -L + \dot{\psi}\frac{\partial L}{\partial\dot{\psi}} + \dot{x}\frac{\partial L}{\partial\dot{x}} = \frac{1}{2}u^2 x + \gamma\frac{\cos\psi}{2\kappa\pi} - \frac{x\Sigma}{\kappa} - \frac{1}{2}\frac{\sin^2\psi}{x}$$

$$H = \frac{x}{2}\left[\dot{\psi}^2 - \frac{\sin^2\psi}{x^2}\right] - \frac{\Sigma}{\kappa}x + \gamma\frac{\cos\psi}{2\pi k}$$

H is conserved because $\frac{\partial L}{\partial s} = 0$. It also leads to the boundary condition for $\gamma(s)$ see eqs 3.7 from [7].

3.1 Physical quantity of the system

The following ones are the physical quantities or the constitutive relations involved in our system of interest:

- R_{pa} particle radius
- $|W|$ adhesive energy density for area unit
- R_{ve} radius of vesicle
- k bending rigidity

We can describe the system using two adimensional quantity:

$$r_{pa} = \frac{R_{pa}}{R_{ve}}$$

$$w = \frac{|W|R_{pa}^2}{\kappa}$$

4 Shape ODE system

The following is the set of differential equation needed to describe the equilibrium shape of a membrane that interacts with an external particle. This system of equation is:

- ordinary because the independent variable is always $\bar{s}$.
- First order because all the derivatives are first derivative
- Non linear because of non linear terms like squares or trigonometric functions.
- it is a system because the dependent variables are coupled.

$$\begin{cases} \frac{d\psi}{ds} = u \\ \frac{du}{ds} = [-\frac{u}{x}\cos\psi + \frac{\cos(\psi)\sin(\psi)}{x^2} + \frac{\gamma\sin\psi}{2\pi\kappa x}] \\ \frac{d\gamma}{ds} = [\pi\kappa(u^2 - \frac{\sin^2\psi}{x^2}) + 2\pi\Sigma] \\ \frac{dx}{ds} = \cos\psi \end{cases} \tag{2}$$

4.1 Initial conditions and parameters

This ODE problem is not in the standard form of initial value problem (IVP) where we know the starting conditions for every variable. The system is even not a complete boundary value problem (BVP) because we dont know the values of some variable at the boundaries. There are several methods to solve BVPs, we will use shooting method.

Boundary conditions We know that at the South Pole ($s = 0$):

$$\psi(s = 0) = 0 \;\; \gamma(s = 0) = 0 \;\; x(s = 0) = 0$$

At the North Pole $s = s^\star$:

$$\bar{\psi}(s^\star) = \bar{\psi}^\star = \pi + \phi$$

$$\bar{x}(s^\star) = \bar{x}^\star = \frac{R_{pa}}{R_{ve}} \sin\phi = r_{pa} \sin\phi$$

$$\bar{u}(s^\star) = \bar{u}^\star = \frac{R_{ve}}{R_{pa}} = \frac{1}{r_{pa}}$$

$$\bar{A}(s^\star) = A - A_{bo} = \frac{\bar{A} - \bar{A}_{bo}}{4\pi R_{ve}^2}$$

Then, we have 4 unknown parameters:

$$\Sigma,\ s^\star,\ u_0,\ u^\star$$

Boundary values of $\gamma(s)$ Hamiltonian $H(s_0) = 0$ and substituting the value for ψ and x I obtain $\gamma(s_0)\cos(\psi(s_0)) = 0$ and then $\gamma(s_0) = 0$. Assuming that H is conserved because it is an energy, $H(s = s^\star) = 0$:

$$\bar{\gamma}^\star = -2\pi r_{pa} \bar{\Sigma} \tan\phi$$

5 Numerical implementation

Oss: See the code at Gitlab repository

- you will choose a value for the wrapping angle ϕ and you will know the A_{bo}
- now you have everything to calculate the bound free energy E_{bo}
- shooting method for the unbound shape:
  - you will start with choosing random free parameters $\Sigma,\ s^\star, u^\star$ and u_0
  - you will numerically integrate the ODE system knowing the initial conditions for ψ, u, γ, x
  - after the integration from the South Pole to the contact point, you calculate the residual function $\mathbf{R}$ as the error between the boundary conditions at $s = s^\star$ and the numerical solution.

    For a fully constrained model (where we track the area A_{tot}), we have 4 unknown free parameters to guess: $\Omega, \bar{\Sigma}, u_0$, and $u_{contact}$. Consequently, we need 4 residuals to have a well-defined boundary value problem. These matching conditions correspond to:

1. The contact angle matches: $\psi(s^\star) - \bar{\psi}^\star = 0$
2. The curvature relates to $u_{contact}$: $u(s^\star) - u_{contact} = 0$
3. The radial coordinate matches: $x(s^\star) - \bar{x}^\star = 0$
4. The total targeted area constraint is satisfied: $A_{tot} - A^\star = 0$

Note on $\gamma^\star$: The Lagrange multiplier γ is *not* a numerical residual. Instead, its value at the contact point ($\bar{\gamma}^\star$) is fixed algebraically by the Hamiltonian conservation law.

Minimizing this 4-dimensional error vector $\mathbf{R}$ corresponds to finding the best parameters that match the numerical solution with the boundary conditions. To minimize this function, we use the non-linear least squares method.

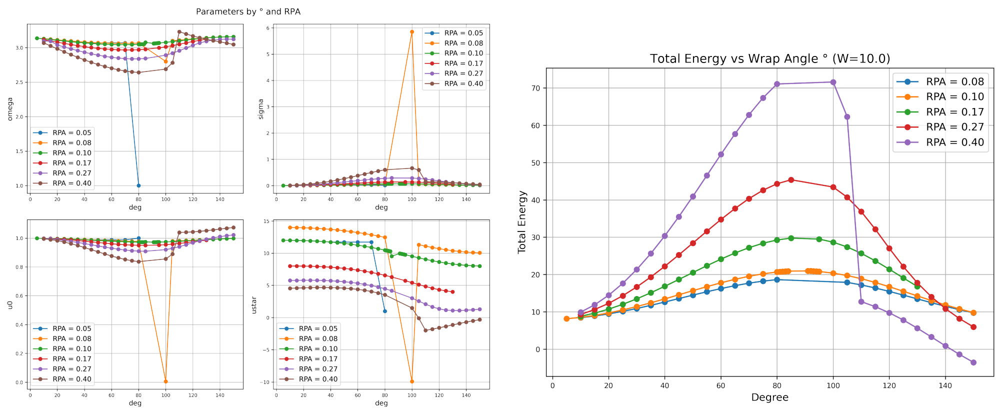
- calculate the total free energy of the system
- estabilish if the total free energy is a minimum/stable state for the chosen combination of constitutive relations and contact angle ϕ.

5.1 Multiple Shooting Method

The integration over the entire domain $[s_0, s^\star]$ can suffer from exponential error growth, causing numerical instabilities. To mitigate this issue, our preferred methodology relies on a **Multiple Shooting Method** optimized with Numba for performance and robustness. Instead of integrating the ODEs in a single pass, the interval is divided into M sub-intervals or segments, bordered by intermediate nodes.
At each step, we calculate a residual function based on the boundary conditions. The residual evaluates the discrepancies at the contact point and continuity at internal nodes between segments:

$$\mathbf{R}_i = \mathbf{Y}_{left}(s_i) - \mathbf{Y}_{right}(s_i)$$

where $\mathbf{Y}$ represents the state vector. Specifically, the matching residuals that are minimized are the differences corresponding to ψ, u, x, and the total vesicle dimensionless area: $A_{tot} - A^\star$.
This ensures stability and convergence across a broad parameter space of wrapping angles ϕ and relative particle sizes r_{pa}.



(a) Optimized parameters landscape.

(b) Total energy ($W = 10.0$).

Figure 3: Results of the Multiple Shooting integration algorithm over various wrapping angles ϕ and dimensionless particle radii r_{pa}.

Note: it is (very) difficult to find a good parameters guess. One way to approach the problem is to try using neutral default params initialization (like $1.0, 1.0, 1.0, 1.0$) and to perform the integration for a various array of contact angles and particle radius. Most of them will be bogus, but a few might be looking decent. The next step is to use the optimized parameters from the decent ones and slowly building all the combinations. Example: you found a good solution for $r_{pa} = 0.1$ and $\phi = 30^o$ just slowly change the particle radius to 0.15, 0.2 using the previous optimized paramters found.

5.2 Expansion at the South pole

At $s = 0$ $x(s)$ goes to zero and it appears as denominator and it leads to divergences causing numerical instability. An easy way to solve this is to regularize the functions at the "South pole" when $s = 0$. Regularization means Taylor expansion of all functions around $s = 0$.
First you need to expand $x(s), \psi(s), u(s), \gamma(s)$ and substitute the expansion in each equation (both sides) of the system. Then equating the coefficient of the polynomials you can get the series coefficient.

$$u(s) = u(0) + u_1 s + \frac{u_2}{2}s^2 + \frac{u_3}{6}s^3 + \frac{u_4}{24}s^4 + O(s^4)$$

$$\cos(\psi(s)) = 1 - \frac{\psi(s)^2}{2} + \frac{\psi(s)^4}{24} = 1 - \frac{1}{2}U_0^2 s^2 + \frac{U_0\psi_2}{2}s^3 + (\frac{\psi_2^2}{8} + \frac{U_0\psi_3}{6})s^4 + O(s^4);$$

Considering the simplest (and not trivial) equation from our system $\frac{dx}{ds} = \cos\psi$ and performing this substitution we'll get the coefficients for x expansion:

$$x_1 = 1,\ x_2 = 0,\ x_3 = -u_0^2,\ x_4 = 0$$

$$x(s) = s - \frac{s^3 u_0^2}{6};$$

The another equations from the system are quite challenging to deal on pen and paper and for that we have used a Mathematica script to perform the expansion.The final values for the coefficient are:

$$x_1 = 1,\ x_2 = 0,\ x_3 = -u_0^2,\ x_4 = 0$$

$$\psi_1 = u_0,\ \psi_2 = 0,\quad \psi_4 = 0$$

$$\psi_3 = \frac{3\gamma_1 u_0}{2\pi k} = \frac{6\pi\Sigma u_0}{2\pi k} = \frac{3\Sigma u_0}{\kappa}$$

$$\gamma_1 = 2\pi\Sigma,\ \gamma_2 = 0$$

$$\gamma_3 = \frac{4}{3}\kappa\pi\psi_3 u_0 = 4\pi(\Sigma u_0) = 4\pi\Sigma u_0,\ \gamma_4 = 0$$

$$A_1 = 0,\ A_2 = 2\pi,\ A_3 = 0,\ A_4 = -2\pi u_0^2$$

Then if you want to reconstruct the series:

$$x(s) = x_1 s + \frac{x_2}{2}s^2 + \frac{x_3}{6}s^3 + \frac{x_4}{24}s^4 = s - \frac{u_0^2}{6}s^3$$

$$\gamma(s) = 2\pi\Sigma s + \frac{2}{9}\kappa\pi\psi_3 u_0 s^3 = 2\pi\Sigma s + \frac{1}{9}u_0(6\pi u_0\Sigma)s^3$$

$$\psi(s) = u_0 s + \frac{3\Sigma u_0}{6\kappa}s^3$$

$$u(s) = u(0) + \frac{u_2 s^2}{2} + \frac{u_3 s^3}{6} + \frac{u_4 s^4}{24} = u_0 + \frac{\psi_3 s^2}{2} + \frac{\psi_4 s^3}{6} + \frac{\psi_5 s^4}{24}$$
$$= u_0 + \frac{\Omega^2}{2}\left(\frac{3\Sigma u_0}{2k}\right) s^2$$

This incorporates the required Ω^2 scale factor from the non-dimensionalization step. The exact mathematical limit without pressure yields an expansion missing the computationally stiff u_0^3 term (often included when working with non-zero pressure systems).
This expansions still need to be non-dimensionalized.

5.3 Expansion at North Pole

Simular studies [4] [3] has expanded the system also at the north pole due to the presence of divergences.
In our problem we will integrate from $s = 0$ to the contact point between particle and vesicle. If the contact point is not too close to the z-axis (rotational simmetry axis) we wont have any divergences. However it is worth checking this condition when $\phi \leq 20°$ or $\phi \geq 165°$ because the contact point became closer.

6 Non-dimensionalization

6.1 Non-dimensionalization step

$$R_{ve} = \sqrt{\frac{A}{4\pi}}$$

Given R_{ve} as length scale, k as basic energy scale and $s^\star$ as the bound arc length we can rewrite our equation in a unitless form:
The dashed symbols are the unitless quantities.

$$\bar{\psi} = \psi \qquad \bar{u} = uR_{ve} \qquad \bar{x} = \frac{x}{R_{ve}}$$

$$\bar{\gamma} = \gamma R_{ve} \quad \bar{\Sigma} = \Sigma\frac{R_{ve}^2}{\kappa} \quad \bar{A} = \frac{A}{4\pi R_{ve}^2}$$

$$\bar{V} = \frac{3V}{4\pi R_{ve}^3} \qquad \bar{s} = \frac{s}{s^\star} \qquad \bar{s^\star} = \frac{s^\star}{s^\star} = 1$$

$$\Omega = \frac{s^\star}{R_{ve}}$$

Substitute in the system we get:

$$\begin{cases} \frac{d\bar{\psi}}{d\bar{s}} = \Omega\bar{u} \\ \frac{d\bar{u}}{d\bar{s}} = \Omega[-\frac{\bar{u}}{\bar{x}}\cos\bar{\psi} + \frac{\cos(\bar{\psi})\sin(\bar{\psi})}{\bar{x}^2} + \frac{\bar{\gamma}\sin\bar{\psi}}{2\pi\kappa\bar{x}}] \\ \frac{d\bar{\gamma}}{d\bar{s}} = \Omega[\pi\kappa(\bar{u}^2 - \frac{\sin^2\bar{\psi}}{\bar{x}^2}) + 2\pi\bar{\Sigma}k] \\ \frac{d\bar{x}}{d\bar{s}} = \Omega\cos\bar{\psi} \end{cases} \tag{3}$$

6.2 Non-dimensionalization of the expansion

The dash symbols are the unitless quantities and we applied the same trasformations previously used.

$$\bar{x_1} = \Omega \qquad \bar{x_3} = -\Omega^3\bar{u_0}^2 \qquad \bar{\psi_1} = \Omega\bar{u_0}$$

$$\bar{\psi_3} = (3\bar{\Sigma}\bar{u_0})\Omega^3 \qquad \bar{u_0} = \bar{u_0} \qquad \bar{u_1} = \frac{\Omega^2}{2}(3\bar{u_0}\bar{\Sigma})$$

$$\gamma_1 = 2\pi\bar{\Sigma}\kappa\Omega \quad \gamma_3 = \frac{4}{3}\pi\kappa\bar{u_0}\Omega^3(3\bar{\Sigma}\bar{u_0}) \quad \bar{V_4} = \frac{9}{2}\bar{u_0}\Omega^4$$

$$\bar{A_2} = \frac{1}{2}\Omega^2 \qquad \bar{A_4} = -\frac{1}{2}\bar{u_0}^2\Omega^4$$

A Jacobian Matrix

This ODE system now is written in the standard form:

$$\frac{d}{d\bar{s}}\mathbf{Y} = \mathbf{f}(\mathbf{Y})$$

where

$$\mathbf{Y} = (\bar{\psi}, \bar{u}, \bar{\gamma}, \bar{x})$$

and

$$\mathbf{f} : \mathbb{R}^4 \to \mathbb{R}^4$$

Oss: Now the following quantities are not dashed but they are already in their adimensional form, so consider for example $\psi(s)$ as $\bar{\psi}(\bar{s})$.
However:

$$\mathbf{J} = \begin{bmatrix} \frac{\partial f_1}{\partial \mathbf{Y}} \\ \frac{\partial f_2}{\partial \mathbf{Y}} \\ \vdots \\ \frac{\partial f_n}{\partial \mathbf{Y}} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial \bar{\psi}} & \frac{\partial f_1}{\partial \bar{u}} & \frac{\partial f_1}{\partial \bar{\gamma}} & \dots & \dots \\ \frac{\partial f_2}{\partial \bar{\psi}} & \frac{\partial f_2}{\partial \bar{u}} & \frac{\partial f_2}{\partial \bar{\gamma}} & & \\ \vdots & \vdots & & & \\ \frac{\partial f_n}{\partial \bar{\psi}} & \frac{\partial f_n}{\partial \bar{u}} & \frac{\partial f_n}{\partial \bar{\gamma}} & \dots & \dots \end{bmatrix}$$

$$\mathbf{J} = \begin{bmatrix} 0 & a_{12} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & a_{24} & 0 & 0 \\ a_{31} & a_{32} & 0 & a_{34} & 0 & 0 \\ a_{41} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & a_{54} & 0 & 0 \\ a_{61} & 0 & 0 & a_{64} & 0 & 0 \end{bmatrix}$$

$$a_{12} = \frac{df_\psi}{du} = \Omega$$

$$a_{21} = \frac{df_u}{d\psi} = \Omega \left(\frac{u}{x} \sin\psi + \frac{1}{x^2} \cos^2\psi + \frac{1}{x^2} \sin^2\psi \frac{\gamma}{2\pi\kappa x} \cos\psi \right)$$

$$a_{22} = \frac{df_u}{du} = -\frac{\Omega}{x} \cos\psi$$

$$a_{23} = \frac{df_u}{d\gamma} = \Omega \frac{\sin\psi}{2\pi\kappa x}$$

$$a_{24} = \frac{df_u}{dx} = \Omega\left(\frac{u}{x^2} \cos\psi - 2\frac{\cos\psi \sin\psi}{x^3} - \frac{\gamma \sin\psi}{2\pi\kappa x^2}\right)$$

$$a_{31} = \frac{df_\gamma}{d\psi} = \Omega(-\pi\kappa\frac{\cos\psi}{x^2})$$

$$a_{32} = \frac{df_\gamma}{du} = \Omega\pi\kappa 2u$$

$$a_{34} = \frac{df_\gamma}{dx} = 2\Omega\pi\kappa\sin\psi\frac{1}{x^3}$$

$$a_{41} = \frac{df_x}{d\psi} = -\Omega\sin\psi$$

$$a_{54} = \frac{df_A}{dx} = \Omega$$

$$a_{61} = \frac{df_V}{d\psi} = \Omega\frac{3}{4}x^2\cos\psi$$

$$a_{64} = \frac{df_V}{dx} = \frac{3}{2}\Omega x\sin\psi$$

B Non-dimensionalization of E_{bo} and E_{un}

$$\bar{E_{bo}} = (-2\pi\bar{W}r_{pa}^2 + 4\pi)(1-\cos\phi)$$

$$\bar{E_{un}} = 2\pi\Omega\kappa\int_0^1\left[\frac{\bar{x}}{2}\left(\bar{u}+\frac{\sin\bar{\psi}}{\bar{x}}\right)^2+\bar{\Sigma}\bar{x}\right]d\bar{s}$$

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